

Networking & Cybersecurity Topics (Categorized)

Networking Fundamentals

1. Internet
 2. MAC Address
 3. IP Addressing (IPv4, IPv6)
 4. Subnetting & Supernetting
 5. CIDR Notation
 6. OSI Model
 7. TCP/IP Model
 8. Network Devices (Hub, Switch, Router, Bridge, Repeater, Gateway, Access Point, Modem)
 9. Synchronous & Asynchronous Data Transmission
 10. Simplex, Half Duplex & Full Duplex Modes
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Core Network Protocols

1. ARP Protocol
 2. ICMP Protocol
 3. IGMP Protocol
 4. QUIC Protocol
 5. Ping Protocol
 6. DNS (Domain Name System)
 7. DHCP (Dynamic Host Configuration Protocol)
 8. FTP, TFTP, SFTP, SCP
 9. SSH & Telnet
 10. SMTP, POP3, IMAP
 11. SNMP (Simple Network Management Protocol)
 12. NTP (Network Time Protocol)
 13. SMB, LDAP, RDP
 14. NAT & PAT (Network & Port Address Translation)
 15. TLS
 16. VPN (SSL, IPsec, PPTP, L2TP)
 17. Firewalls & CDN
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Transport Layer & Web Technologies

1. Well-Known Ports & Their Functions
2. TCP & UDP Protocols (Usage & Data Flow)
3. TCP 3-Way Handshake
4. TCP 4-Way Handshake
5. TCP Flags (SYN, ACK, FIN, RST, PSH, URG)

6. HTTP Methods (GET, POST, PUT, PATCH, TRACE, HEAD, OPTIONS)
 7. Netstat & Traceroute
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Wireless Networking & Security

1. Encryption Protocols (WEP, WPA, WPA2, WPA3)
 2. Wireless Authentication (PSK, Enterprise)
 3. Wireless Attacks (Deauth, Evil Twin, KRACK)
 4. Rogue DHCP & Evil Twin
 5. VLAN Hopping
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Cybersecurity Attacks & Threats

1. ARP Spoofing / Poisoning
 2. DNS Spoofing
 3. IP Spoofing
 4. MAC Flooding
 5. DHCP Starvation
 6. SYN Flood / DoS / DDoS
 7. MITM (Man-in-the-Middle)
 8. Sniffing & Eavesdropping
 9. Replay Attacks
 10. Session Hijacking
 11. ICMP Flood / Ping of Death / Smurf Attack
 12. Routing Attacks
 13. SSL Stripping
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Pentesting Techniques & Post-Exploitation

1. Port Scanning Detection & Prevention
2. Pivoting Through Networks
3. Traffic Redirection & Tunneling
4. Reverse Shells & Bind Shells